(ii) Electro Dispersion:

A brown colloidal solution of platinum was first prepared by George Bredig in 1898. An electrical arc is struck between electrodes dispersed in water surrounded by ice. When a current of 1 amp /100 V is passed an arc produced forms vapours of metal which immediately condense to form colloidal solution. By this method colloidal solution of many metals like copper, silver, gold, platinum, etc. can be prepared Alkali hydroxide is added as an stabilising agent for the colloidal solution.

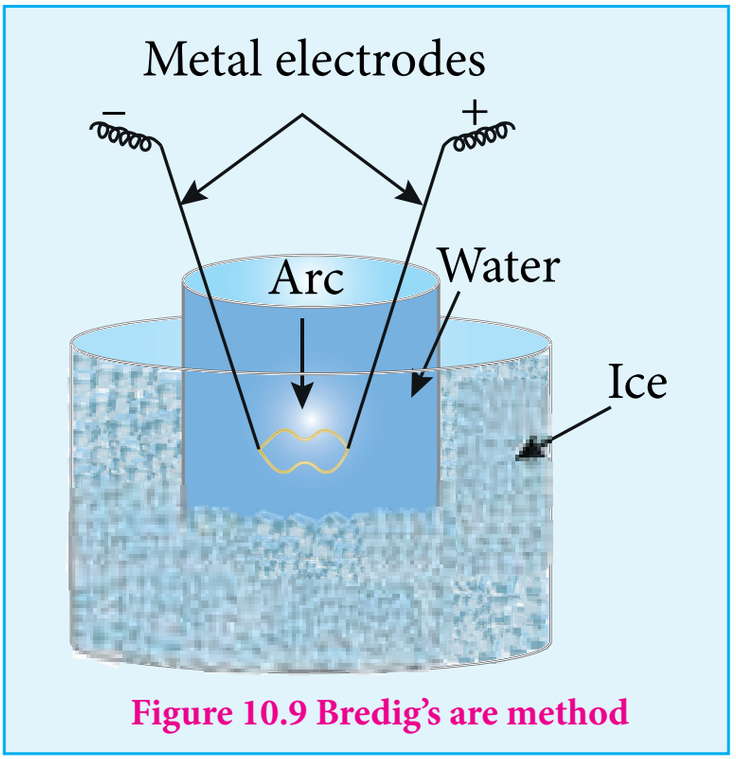


Figure 10.9 Bredig's are method

Svedberg modified this method for the preparation of non aqueous inflammable liquids like pentane, ether and benzene, etc using high frequency alternating current which prevents the decomposition of liquid.

(iii) Ultrasonic dispersion

Sound waves of frequency more than 20kHz (audible limit) could cause transformation of coarse suspension to colloidal dimensions.

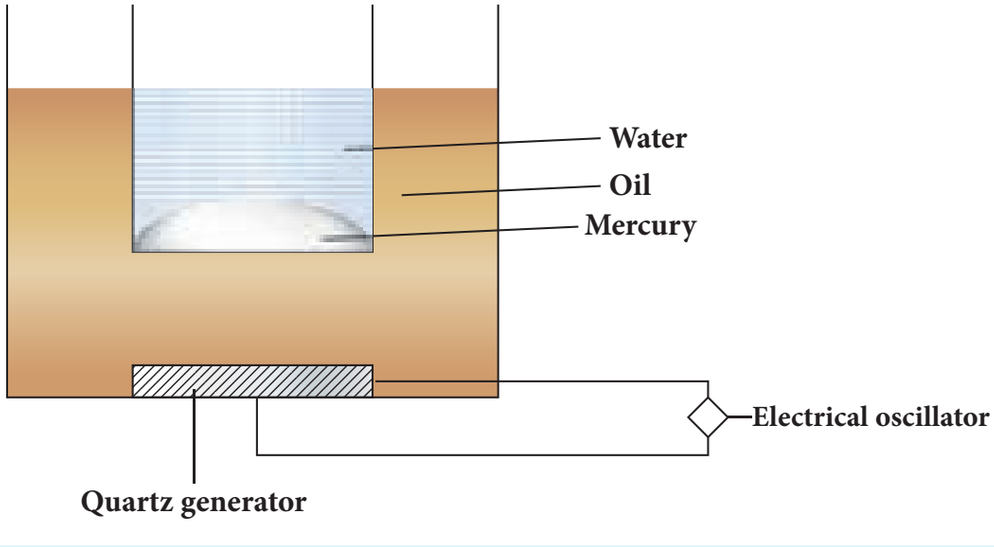


Figure : 10.10 Ultrasonic dispersion

Claus obtained mercury sol by subjecting mercury to sufficiently high frequency ultrasonic vibrations.

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